KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4091 | 4 Credits: 4 | Program Core | Source-grounded

Polymer Processing &

 To understand the different mixing equipments and mixing processes used in rubber

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4091 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4091.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	4091
Course title	Polymer Processing &
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To understand the different mixing equipments and mixing processes used in rubber
- To understand machinery parts of different mixers, moulding equipments, extruder,
- calendar and thermoforming machine.
- To acquire the knowledge of different processing methods for the polymer materials.

Course outcomes

CO	Official outcome	Study level
CO1	Describe the mixing machinery and process. 12 Understanding	See official syllabus
CO2	Explain the moulding process and machinery. 17 Applying Explain the working principles, parts and	See official syllabus
CO3	accessories of extruders and blow moulding 17 Applying machinery Paraphrase rotational moulding, calendars,	See official syllabus
CO4	12 Understanding thermoforming and auxiliary equipments	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe the mixing machinery and process.	4	As specified
Module 2	Explain the moulding process and machinery.	4	As specified
Module 3	blow moulding machinery	4	As specified
Module 4	equipments	3	As specified

MODULE 1

Describe the mixing machinery and process.

This module is presented from the official syllabus scope for Polymer Processing &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the mixing machinery and process.
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

Classify different mixing equipments 1 Remembering Explain the working principle, parts and

Classify different mixing equipments 1 Remembering Explain the working principle, parts and is an official topic in Module 1 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify different mixing equipments 1 Remembering Explain the working principle, parts and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify different mixing equipments 1 remembering explain the working principle, parts and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify different mixing equipments 1 Remembering Explain the working principle, parts and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Classify different mixing equipments 1 Remembering Explain the working principle, parts and □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Understanding mixing sequences of two roll. 3 Explain the working principle, parts and mixing

Understanding mixing sequences of two roll. 3 Explain the working principle, parts and mixing is an official topic in Module 1 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding mixing sequences of two roll. 3 Explain the working principle, parts and mixing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding mixing sequences of two roll. 3 explain the working principle, parts and mixing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding mixing sequences of two roll. 3 Explain the working principle, parts and mixing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Understanding mixing sequences of two roll. 3 Explain the working principle, parts and mixing □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding sequences of internal mixers. Illustrate Continuous mixing equipments and

4 Understanding sequences of internal mixers. Illustrate Continuous mixing equipments and is an official topic in Module 1 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding sequences of internal mixers. Illustrate Continuous mixing equipments and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding sequences of internal mixers. illustrate continuous mixing equipments and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding sequences of internal mixers. Illustrate Continuous mixing equipments and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding sequences of internal mixers. Illustrate Continuous mixing equipments and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Understanding plastic compounding equipments. 4 Mixing machinery Classification- batch and continuous. Working principle- parts- accessories, safety devices-uses of two roll mill and internal mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding.

Understanding plastic compounding equipments. 4 Mixing machinery Classification- batch and continuous. Working principle- parts- accessories, safety devices-uses of two roll mill and internal mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding. is an official topic in Module 1 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding plastic compounding equipments. 4 Mixing machinery Classification- batch and continuous. Working principle- parts- accessories, safety devices-uses of two roll mill and internal mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding plastic compounding equipments. 4 mixing machinery classification- batch and continuous. working principle- parts- accessories, safety devices-uses of two roll mill and internal mixers. differentiate banbury and intermix. fill factor. mixing sequences-distributive mixing, dispersive mixing- upside down mixing. continuous mixing - farrel continuous mixer. auxiliary mixing equipments used for plastic compounding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding plastic compounding equipments. 4 Mixing machinery Classification- batch and continuous. Working principle- parts- accessories, safety devices-uses of two roll mill and internal mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Understanding plastic compounding equipments. 4 Mixing machinery Classification- batch and continuous. Working principle- parts- accessories, safety devices-uses of two roll mill and internal mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe the mixing machinery and process.

  M1.01       Classify different mixing equipments                  1
Remembering


              Explain the working principle, parts and
  M1.02
Understanding
              mixing sequences of two roll.                         3
              Explain the working principle, parts and mixing
  M1.03                                                             4
Understanding
              sequences of internal mixers.
              Illustrate Continuous mixing equipments and
  M1.04
Understanding
              plastic compounding equipments.                       4

                                         Contents:
Mixing machinery Classification- batch and continuous. Working principle- parts-
accessories, safety devices-uses of two roll mill and internal mixers.
Differentiate Banbury
and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive
mixing- upside
down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing
equipments
used for plastic compounding.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the moulding process and machinery.

This module is presented from the official syllabus scope for Polymer Processing &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the moulding process and machinery.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

6 Understanding moulding machine and process. Explain the injection moulding machine and 7

6 Understanding moulding machine and process. Explain the injection moulding machine and 7 is an official topic in Module 2 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding moulding machine and process. Explain the injection moulding machine and 7 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding moulding machine and process. explain the injection moulding machine and 7 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding moulding machine and process. Explain the injection moulding machine and 7 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 6 Understanding moulding machine and process. Explain the injection moulding machine and 7 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Understanding process Identify different defects and Troubleshooting

Understanding process Identify different defects and Troubleshooting is an official topic in Module 2 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding process Identify different defects and Troubleshooting using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding process identify different defects and troubleshooting should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding process Identify different defects and Troubleshooting means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Understanding process Identify different defects and Troubleshooting □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Applying of I/M, C/M and T/M process

2 Applying of I/M, C/M and T/M process is an official topic in Module 2 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying of I/M, C/M and T/M process using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying of i/m, c/m and t/m process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying of I/M, C/M and T/M process means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Applying of I/M, C/M and T/M process [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe RIM and RRIM 2 Understanding Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding. Generalize the working principles, parts and accessories of extruders and

Describe RIM and RRIM 2 Understanding Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding. Generalize the working principles, parts and accessories of extruders and is an official topic in Module 2 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe RIM and RRIM 2 Understanding Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding. Generalize the working principles, parts and accessories of extruders and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe rim and rrim 2 understanding compression moulding - process - moulds - flash, positive - semi positive. hydraulic press - working principle-parts-accessories. transfer moulding - process, machinery. injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. differentiate rubber and plastic injection moulding. generalize the working principles, parts and accessories of extruders and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe RIM and RRIM 2 Understanding Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding. Generalize the working principles, parts and accessories of extruders and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe RIM and RRIM 2 Understanding Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding. Generalize the working principles, parts and accessories of extruders and □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the moulding process and machinery.

            Discuss the compression and transfer
 M2.01                                                                6
Understanding
            moulding machine and process.
            Explain the injection moulding machine and                7
 M2.02
Understanding
            process
            Identify different defects and Troubleshooting
 M2.03                                                                2
Applying
            of I/M, C/M and T/M process
 M2.04      Describe RIM and RRIM                                     2
Understanding

            Series Test – I                                           1

Contents:
Compression moulding - process - moulds - flash, positive - semi positive.
Hydraulic
press - working principle-parts-accessories. Transfer moulding - process,
machinery.
Injection moulding- classifications –for rubbers and plastics, ram &
reciprocating screw
type. Reciprocating screw type - parts - accessories - moulds - clamping system-
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

blow moulding machinery

This module is presented from the official syllabus scope for Polymer Processing &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: blow moulding machinery
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate the parts and accessories of extruders.

Illustrate the parts and accessories of extruders. is an official topic in Module 3 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the parts and accessories of extruders. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the parts and accessories of extruders. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the parts and accessories of extruders. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Illustrate the parts and accessories of extruders. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Illustrate working process of extruders.

Illustrate working process of extruders. is an official topic in Module 3 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate working process of extruders. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate working process of extruders. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate working process of extruders. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Illustrate working process of extruders. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Categorize different types of extruders.

Categorize different types of extruders. is an official topic in Module 3 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Categorize different types of extruders. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, categorize different types of extruders. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Categorize different types of extruders. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Categorize different types of extruders. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Explain blow moulding process 4 Understanding Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding- parison programming - injection blow moulding. Paraphrase rotational moulding, calendars, thermoforming and auxiliary

Explain blow moulding process 4 Understanding Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding- parison programming - injection blow moulding. Paraphrase rotational moulding, calendars, thermoforming and auxiliary is an official topic in Module 3 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain blow moulding process 4 Understanding Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding- parison programming - injection blow moulding. Paraphrase rotational moulding, calendars, thermoforming and auxiliary using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain blow moulding process 4 understanding extrusion - definition - process. extruder – classifications - extruder size - cold feed - hot feed - t - head - dual - twin screw - vented. extruder –screw type - parts, accessories – blown film extrusion. blow moulding - extrusion blow moulding- parison programming - injection blow moulding. paraphrase rotational moulding, calendars, thermoforming and auxiliary should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain blow moulding process 4 Understanding Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding- parison programming - injection blow moulding. Paraphrase rotational moulding, calendars, thermoforming and auxiliary means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain blow moulding process 4 Understanding Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding- parison programming - injection blow moulding. Paraphrase rotational moulding, calendars, thermoforming and auxiliary □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
blow moulding machinery

 M3.01      Illustrate the parts and accessories of extruders.        5
Understanding

 M3.02      Illustrate working process of extruders.                  5
Understanding

 M3.03      Categorize different types of extruders.                  3
Understanding

 M3.04      Explain blow moulding process                             4
Understanding

Contents:
Extrusion - Definition - process. Extruder – classifications - extruder size - cold
feed - hot
feed - T - head - dual - twin screw - vented. Extruder –screw type - parts,
accessories –
blown film extrusion. Blow moulding - extrusion blow moulding- parison programming
-
injection blow moulding.

            Paraphrase rotational moulding, calendars, thermoforming and auxiliary
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

equipments

This module is presented from the official syllabus scope for Polymer Processing &. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: equipments
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Illustrate calendaring process and machinery

Illustrate calendaring process and machinery is an official topic in Module 4 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate calendaring process and machinery using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate calendaring process and machinery should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate calendaring process and machinery means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Illustrate calendaring process and machinery [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe various thermoforming process

Describe various thermoforming process is an official topic in Module 4 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe various thermoforming process using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe various thermoforming process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe various thermoforming process means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Describe various thermoforming process [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain rotational moulding process 4 Understanding Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process. Text / Reference: T/R Book Title/Author Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth T1 Hainemann Ltd, 1982) Rubber processing and production organization -Philip K Freakley (Publisher: R1 Springer 2011). R2 Blow moulding -David A Jenners R3 Injection moulding -Irvin Ruhin Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing R4 Co. 1986) Online Resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com

Explain rotational moulding process 4 Understanding Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process. Text / Reference: T/R Book Title/Author Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth T1 Hainemann Ltd, 1982) Rubber processing and production organization -Philip K Freakley (Publisher: R1 Springer 2011). R2 Blow moulding - David A Jenners R3 Injection moulding -Irvin Ruhin Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing R4 Co. 1986) Online Resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com is an official topic in Module 4 of Polymer Processing &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain rotational moulding process 4 Understanding Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process. Text /

Reference: T/R Book Title/Author Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth T1 Hainemann Ltd, 1982) Rubber processing and production organization -Philip K Freakley (Publisher: R1 Springer 2011). R2 Blow moulding -David A Jenners R3 Injection moulding -Irvin Ruhin Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing R4 Co. 1986) Online Resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain rotational moulding process 4 understanding calendar - configuration, parts, accessories. roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. temperature control system of calendars. thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. rotational moulding - process. text / reference: t/r book title/author rubber technology and manufacture-c.m. blow (publisher – butterworth t1 hainemann ltd, 1982) rubber processing and production organization -philip k freakley (publisher: r1 springer 2011). r2 blow moulding -david a jenners r3 injection moulding - irvin ruhin plastic technology hand book - william j patton (publisher - reston publishing r4 co. 1986) online resources: sl.no website link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain rotational moulding process 4 Understanding Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process. Text / Reference: T/R Book Title/Author Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth T1 Hainemann Ltd, 1982) Rubber processing and production organization -Philip K Freakley (Publisher: R1 Springer 2011). R2 Blow moulding -David A Jenners R3 Injection moulding - Irvin Ruhin Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing R4 Co. 1986) Online Resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain rotational moulding process 4 Understanding Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process. Text / Reference: T/R Book Title/Author Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth T1 Hainemann Ltd, 1982) Rubber processing and production organization -Philip K Freakley (Publisher: R1 Springer 2011). R2 Blow moulding -David A Jenners R3 Injection moulding -Irvin Ruhin Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing R4 Co. 1986) Online Resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
equipments
 M4.01      Illustrate calendaring process and machinery              4
Understanding

 M4.02      Describe various thermoforming process                    4
Understanding

 M4.03      Explain rotational moulding process                       4
Understanding

            Series Test – II                                          1
Contents:
Calendar - configuration, parts, accessories. Roll deflection and compensation
technique
like roll cambering (crowing) roll crossing and roll bending. Temperature control
system of
calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted
forming -
products - examples. Rotational moulding - process.

Text / Reference:

  T/R                                  Book Title/Author
         Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth
  T1
         Hainemann Ltd, 1982)
         Rubber processing and production organization -Philip K Freakley
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Classify different mixing equipments 1 Remembering Explain the working principle, parts and. (3 marks)

Answer: Begin with the standard definition or identity of *Classify different mixing equipments 1 Remembering Explain the working principle, parts and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understanding mixing sequences of two roll. 3 Explain the working principle, parts and mixing. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding mixing sequences of two roll. 3 Explain the working principle, parts and mixing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding sequences of internal mixers. Illustrate Continuous mixing equipments and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding sequences of internal mixers. Illustrate Continuous mixing equipments and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understanding plastic compounding equipments. 4 Mixing machinery Classification- batch and continuous. Working principle- parts- accessories, safety devices-uses of two roll mill and internal mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding.. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding plastic compounding equipments. 4 Mixing machinery Classification- batch and continuous. Working principle- parts- accessories, safety devices-uses of two roll mill and internal* *mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 6 Understanding moulding machine and process. Explain the injection moulding machine and 7. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding moulding machine and process. Explain the injection moulding machine and 7*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understanding process Identify different defects and Troubleshooting. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding process Identify different defects and Troubleshooting*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Applying of I/M, C/M and T/M process. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying of I/M, C/M and T/M process*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe RIM and RRIM 2 Understanding Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding. Generalize the working principles, parts and accessories of extruders and. (3 marks)

Answer: Begin with the standard definition or identity of *Describe RIM and RRIM 2 Understanding Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding. Generalize the working principles, parts and accessories of extruders and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Illustrate the parts and accessories of extruders.. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the parts and accessories of extruders.*. State the governing principle, list the key components or stages, or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate working process of extruders.. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate working process of extruders.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Categorize different types of extruders.. (3 marks)

Answer: Begin with the standard definition or identity of *Categorize different types of extruders.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain blow moulding process 4 Understanding Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding-parison programming - injection blow moulding. Paraphrase rotational moulding, calendars, thermoforming and auxiliary. (3 marks)

Answer: Begin with the standard definition or identity of *Explain blow moulding process 4 Understanding Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding-*

parison programming - injection blow moulding. Paraphrase rotational moulding, calendars, thermoforming and auxiliary. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Illustrate calendaring process and machinery. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate calendaring process and machinery*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe various thermoforming process. (3 marks)

Answer: Begin with the standard definition or identity of *Describe various thermoforming process*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain rotational moulding process 4 Understanding Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process. Text / Reference: T/R Book Title/Author Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth T1 Hainemann Ltd, 1982) Rubber processing and production organization -Philip K Freakley (Publisher:

R1 Springer 2011). R2 Blow moulding -David A Jenners R3 Injection moulding -Irvin Ruhin Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing R4 Co. 1986) Online Resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com. (3 marks)

Answer: Begin with the standard definition or identity of *Explain rotational moulding process 4 Understanding Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process. Text / Reference: T/R Book Title/Author Rubber Technology and manufacture- C.M. Blow (Publisher – Butterworth T1 Hainemann Ltd, 1982) Rubber processing and production organization -Philip K Freakley (Publisher: R1 Springer 2011). R2 Blow moulding -David A Jenners R3 Injection moulding -Irvin Ruhin Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing R4 Co. 1986) Online Resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in 4 www.sciencedirect.com 5 www.timcorubber.com*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Classify different mixing equipments 1 Remembering Explain the working principle, parts and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **6 Understanding moulding machine and process. Explain the injection moulding machine and 7**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the parts and accessories of extruders.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Illustrate calendaring process and machinery**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

IN-PAGE SEARCH

Search this lesson